Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE
Course Code & Title: CCS339 Cryptocurrency and Blockchain technologies
Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit II / Block propagation and block relay
Course Outcome: CO 2

Unit Learning Outcome: UO 2b

Activity Chosen: Learning by Teaching

Justification:

The main objective of the learning by teaching activity was to promote active learning and knowledge consolidation among the students. By assigning topics to the students and having them teach those topics to their peers, the activity aimed to reinforce their understanding, improve communication skills, and foster a collaborative learning environment.

- **Time Allotted for the Activity:** 35 minutes

Details of the Implementation:

- Prior to the session, the instructor asked the students about willingness about teaching to the particular topic.
- Shanmugalakshmi P, Gomathi Lakshmi M shown interest to take class.
- Student was assigned a topic Block propagation and block relay based on their relevance and complexity.
- Student was given time to prepare their assigned topics. she was encouraged to gather information from various sources, organize the content effectively, and plan their teaching approach.
- During the session, students took turns presenting their assigned topics to their classmates.
- After that, a question-and-answer session was held to encourage active participation and clarify any doubts.
- This allowed students to engage in discussions and exchange additional insights related to the topic.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PSO1
CO 2	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PSO1**
	1	1	1	1	1
Justification for correlation	Students will be able to understand the fundamental concept of Block propagation and block relay by applying the knowledge of mathematics.	Students will analyze blockchain use cases and scenarios using engineering sciences.	Students will gain proficiency in writing, deploying, and interacting with smart contracts	Students will be able to grasp the concept of peer-to-peer (P2P) networks by investigate complex problems	Students will be able to apply their knowledge of Bitcoin Scripts to excel in software development.

- **Images / Screenshot of the practice:**

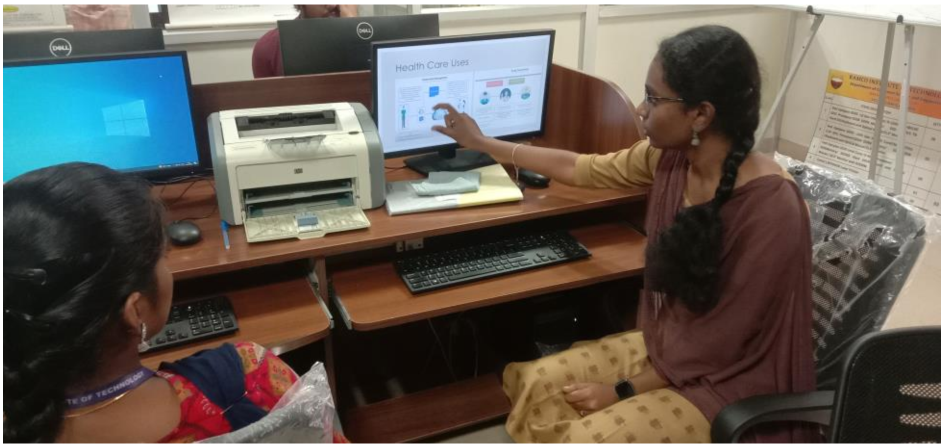

Figure 1: Discussion about **block relay** *by Gomathi Lakshmi M*

Figure 2: Discussion about Block propagation by Shanmugalakshmi P

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***
  - The activity encouraged to have a deeper understanding of the subject matter and improve his retention of the information.
  - Students felt they are able to understand the topic.

❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

  - Students spent their time in self-learning.
  - This activity encouraged the students to share their knowledge with others.

❖ ***Challenges faced in implementation:***
  - Due to shyness, some students hesitate to teach.
  - The instructor did not have sufficient time to discuss the activity after completion with the student.

References:

1. https://effectiviology.com/protege-effect-learn-by-teaching/
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5_VA_CS3362_Learning%20by%20teaching.pdf
4. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/Unit_5_Learning_by_Teaching.pdf

Signature of Faculty Member **HOD**